

Student notes

# What makes working code an engineered system?

Student notes for lecture 1. Every scene shows a situation, asks you to predict, reveals the mechanism, changes one condition, compares the outcomes and names the principle. The pages that follow keep the selected states of each scene with their captions.

Scenes in this lecture

| #  | SCENE (STUDENT HEADING)                                  | ROLE   | DEEP LINK                                               |
|----|----------------------------------------------------------|--------|---------------------------------------------------------|
| 01 | Two green confirmations cannot both be right             | Anchor | <a href="#">week-01.html#/two-confirmations</a>         |
| 02 | The Reserve button is not the whole system               | Bridge | <a href="#">week-01.html#/system-boundary</a>           |
| 03 | The same screen can serve one person and exclude another | Anchor | <a href="#">week-01.html#/stakeholder-lenses</a>        |
| 04 | Faster is not a complete description of better           | Bridge | <a href="#">week-01.html#/quality-tradeoffs</a>         |
| 05 | A failure can teach the next requirement                 | Anchor | <a href="#">week-01.html#/feedback-lifecycle</a>        |
| 06 | What would make you believe this works?                  | Bridge | <a href="#">week-01.html#/evidence-not-confidence</a>   |
| 07 | Who carries the cost of an engineering shortcut?         | Bridge | <a href="#">week-01.html#/responsibility-and-impact</a> |
| 08 | One domain, many kinds of explanation                    | Bridge | <a href="#">week-01.html#/course-evidence-map</a>       |

Campus Rooms is a fictional room-booking service; every name, number and incident is a teaching example unless a source is named. Values marked example assumption are illustrative, not measurements.

## The screen says **Confirmed**. Is that enough?

Campus Rooms is a fictional service that reserves campus rooms. Every name, number and incident in this course is a teaching example.

### The situation

Ari (U-01) reserves room C101 for 10:00–11:00. The screen answers:  
Confirmed: C101, 10:00–11:00. Your booking is saved. A confirmation message is on its way.

### The question of this week

What would you need to see before you trust this message? (A) The green tick and the word Confirmed. (B) The stored calendar for the room. (C) What happens at the door at 10:00.

### About the examples

Campus Rooms is a fictional room-booking service. Every name, number and incident in this course is a teaching example unless a source is named on the page. Room C101 and the 10:00 to 11:00 hour recur through the whole course.

## What you already know, and where this lecture goes

What makes working code an engineered system?

### Start from a program you know

Think of a program you have written or used every week. What does it do when everything goes as planned? What could make it fail outside that happy path: a second user, a lost connection, a rule that changes, a wrong assumption? Who else is affected when it fails?

### Where this applies

A throwaway calculation on your own machine needs none of this. A shared booking service that other people depend on needs all of it. The care you take grows with the consequences of a failure and with how much you do not know yet.

### The thread that runs through the course

Every habit in this course, a precise rule, an independent check, a bounded permission, a reviewable change, is also what it takes to direct and check an automated engineering assistant. Weeks 2, 7, 8, 9, 10 and 12 each return to that for a moment; Week 13 looks at it in full.

### By the end of this lecture you can

- 1 Distinguish a locally correct function from a dependable user-facing service.
- 2 Identify stakeholders, system boundaries, and competing quality concerns.
- 3 Explain why delivery and maintenance belong in the engineering lifecycle.
- 4 Separate an observation, an assumption, and evidence supporting a claim.
- 5 Explain one technical decision in terms of benefit, cost, and affected people.

## Two green confirmations cannot both be right

### Panel 1 of 3 · Two confirmations on screen, two overlapping bookings stored

Condition: Baseline: each screen checks its own copy of the room calendar and decides alone    State: 5 of 7

The stored calendar holds **two confirmed bookings for C101 in the same hour**. Rule R-01 says a room has no overlapping confirmed bookings, so the stored record is wrong even though each screen looked right.

The diagram illustrates a race condition in a room booking system. It shows two user interfaces, Ari (U-01) and Bo (U-02), both displaying a 'Confirmed' status for a booking in room C101 from 10:00 to 11:00. The central 'Stored calendar' shows two overlapping confirmed bookings for the same room and time, one from Ari and one from Bo. A red banner at the bottom states 'R-01 broken: two confirmed bookings overlap'.

**Ari · U-01**

ROOM: C101

TIME: 10:00–11:00

CALENDAR COPY: 10:00 free

Reserve

Confirmed  
C101, 10:00–11:00

**Stored calendar · C101 · today**

09:00

10:00

11:00

12:00

13:00

R-01 broken: two confirmed bookings overlap

**Bo · U-02**

ROOM: C101

TIME: 10:00–11:00

CALENDAR COPY: 10:00 free

Reserve

Confirmed  
C101, 10:00–11:00

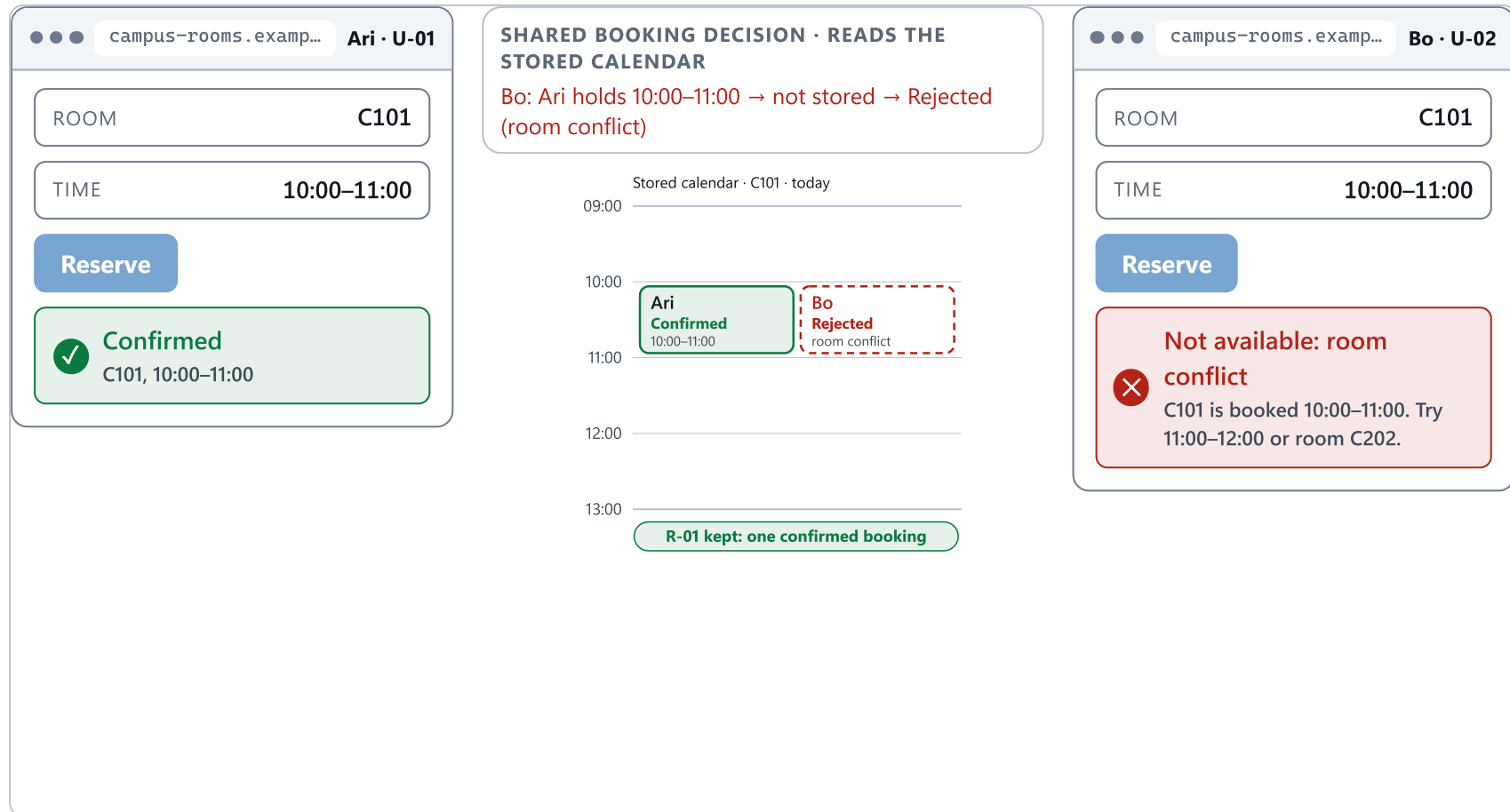
Rule R-01, no overlapping confirmed bookings for the same room, is broken in the stored calendar although each screen showed success. Each screen decided from its own copy of the calendar, and Bo's copy was stale.

## Two green confirmations cannot both be right

### Panel 2 of 3 · One shared decision: one confirmation, one clear rejection

Condition: Changed condition: one shared decision reads the stored calendar before any screen says Confirmed    State: 3 of 5

Bo clicks Reserve. The shared decision finds Ari's booking in the same hour and answers **not available: room conflict**. Bo's screen shows the rejection and offers 11:00–12:00 or room C202.



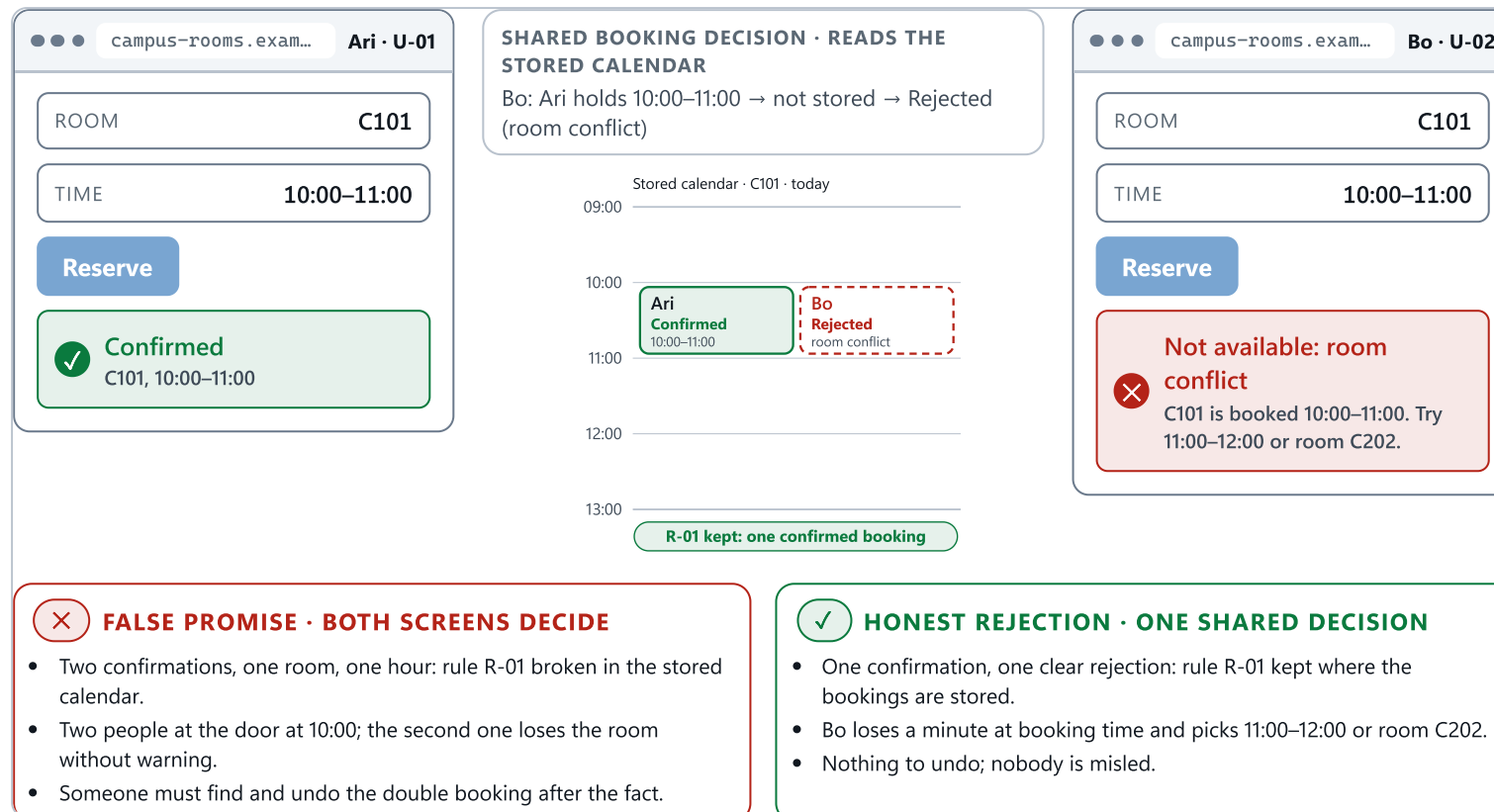
Rule R-01 is kept: the shared decision reads the stored calendar before any screen says Confirmed. Bo sees the reason and two alternatives.

## Two green confirmations cannot both be right

### Panel 3 of 3 · Compare the consequences

Condition: **Changed condition: one shared decision reads the stored calendar before any screen says Confirmed** State: 4 of 5

Compare the two outcomes. A false promise sends two people to one door. An honest rejection costs Bo a minute at booking time and misleads nobody. How the shared decision stays correct when both requests arrive in the same instant is the question for Week 6.



How the shared decision stays correct when both requests arrive in the same instant is the question for Week 6.

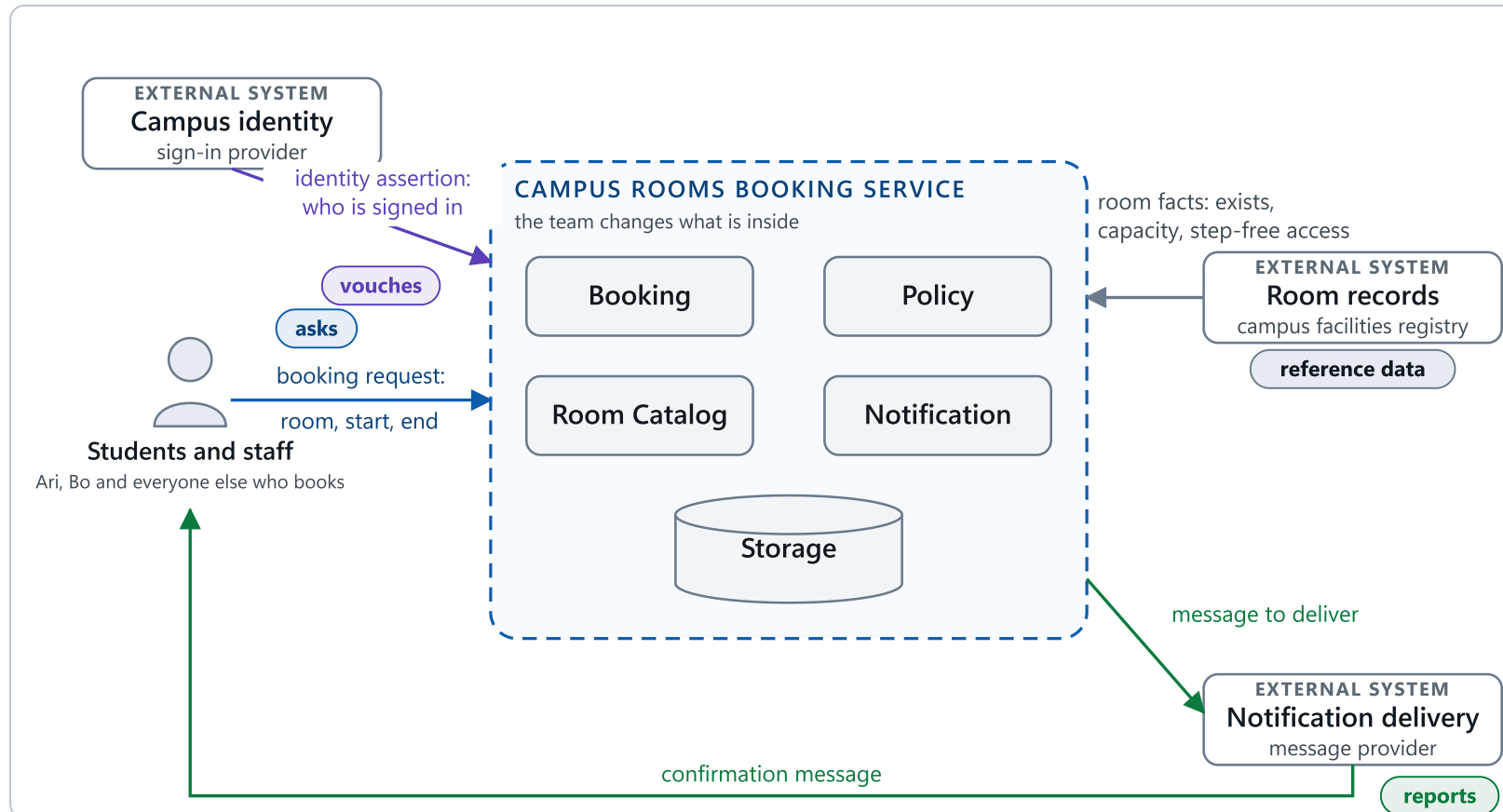
**Principle · Engineering judges behavior against an intended outcome and its consequences, not appearance alone.**

# The Reserve button is not the whole system

## Panel 1 of 2 · The context map with every interaction labeled

Condition: **Baseline:** every external system available    State: 6 of 7

A booking request, an identity assertion and a notification are three kinds of information. One **asks**, one **vouches** for a fact, one **reports** a result. The service treats them differently: it decides on the request, trusts the assertion within limits, and hands the report to a provider.



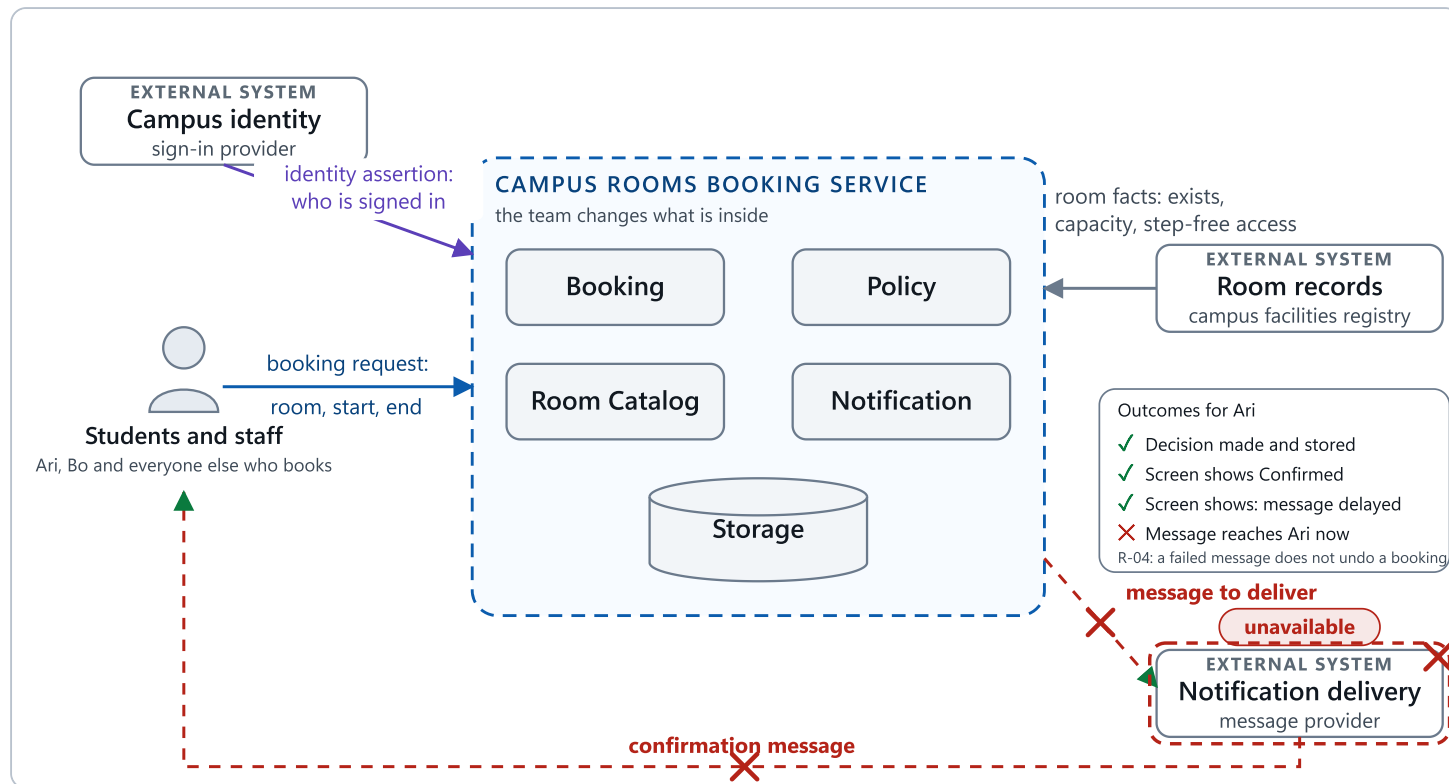
Inside the dashed boundary: Booking, Policy, Room Catalog, Notification and Storage, the modules the team changes. Outside: students and staff, campus identity, room records and notification delivery.

## The Reserve button is not the whole system

### Panel 2 of 2 · One dependency unavailable: what remains possible

Condition: Changed condition: notification delivery unavailable; room storage unchanged    State: 2 of 3

Still possible: the decision is made and stored; Ari's screen shows Confirmed and says that the message is delayed. Not possible right now: the message reaching Ari. Rule R-04: a failed notification does not undo a confirmed booking, and the pending message stays visible and recoverable.



Notification delivery is marked with a broken line and the word unavailable. Still possible: The booking decision is made and stored; Ari's screen shows Confirmed; Ari's screen shows that the confirmation message is delayed. Not possible now: The confirmation message reaches Ari now (rule R-04).

**Principle** · A system boundary identifies responsibility and dependencies; it does not make external behavior disappear.

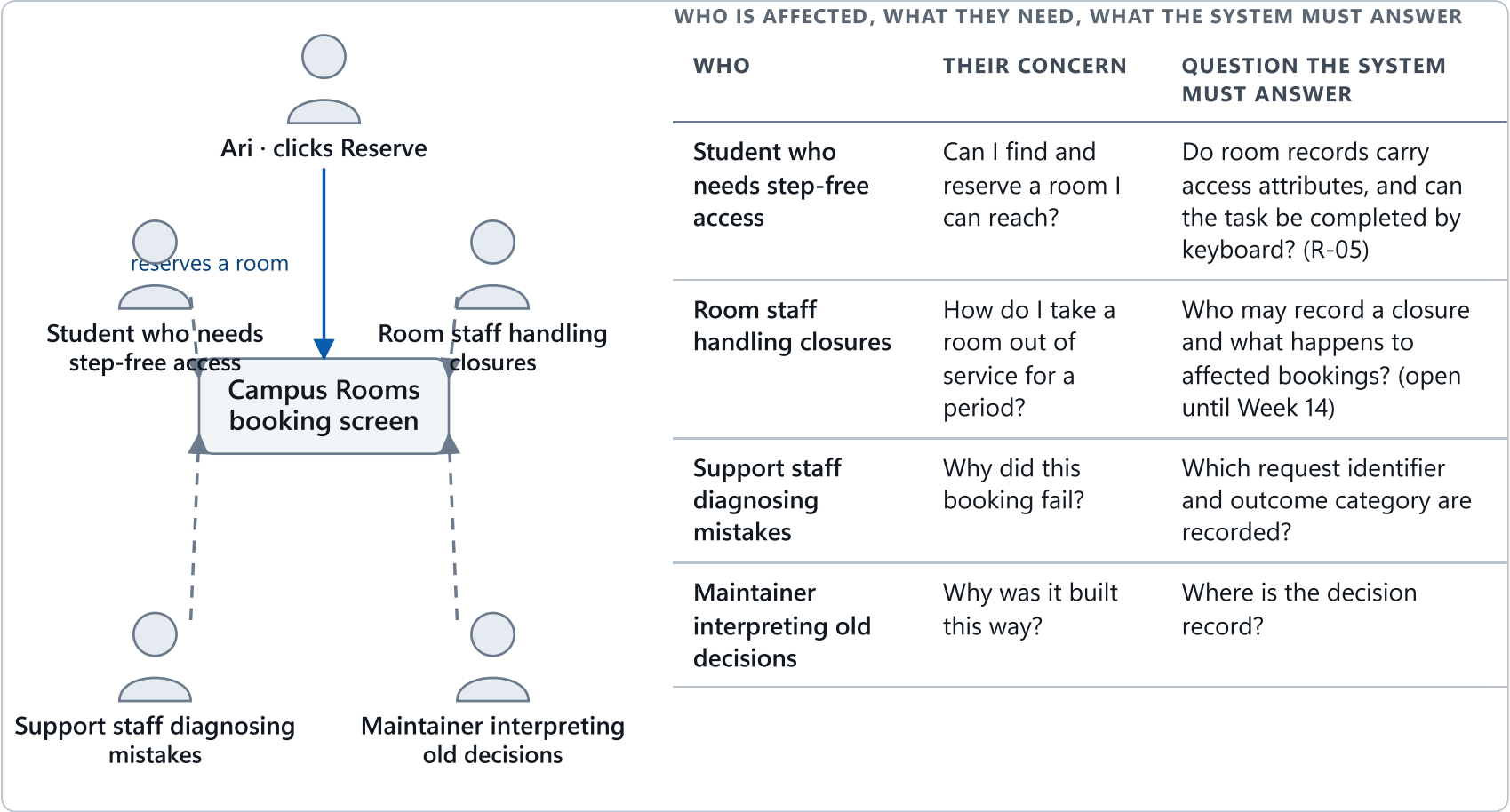


The same screen can serve one person and exclude another

Panel 1 of 2 · Who is affected and what the system must answer

Condition: Baseline: the polished booking screen, used with a mouse    State: 6 of 7

Each concern became a question the system must answer: about room data, closures, records and decisions. Room closures stay an open question until Week 14.



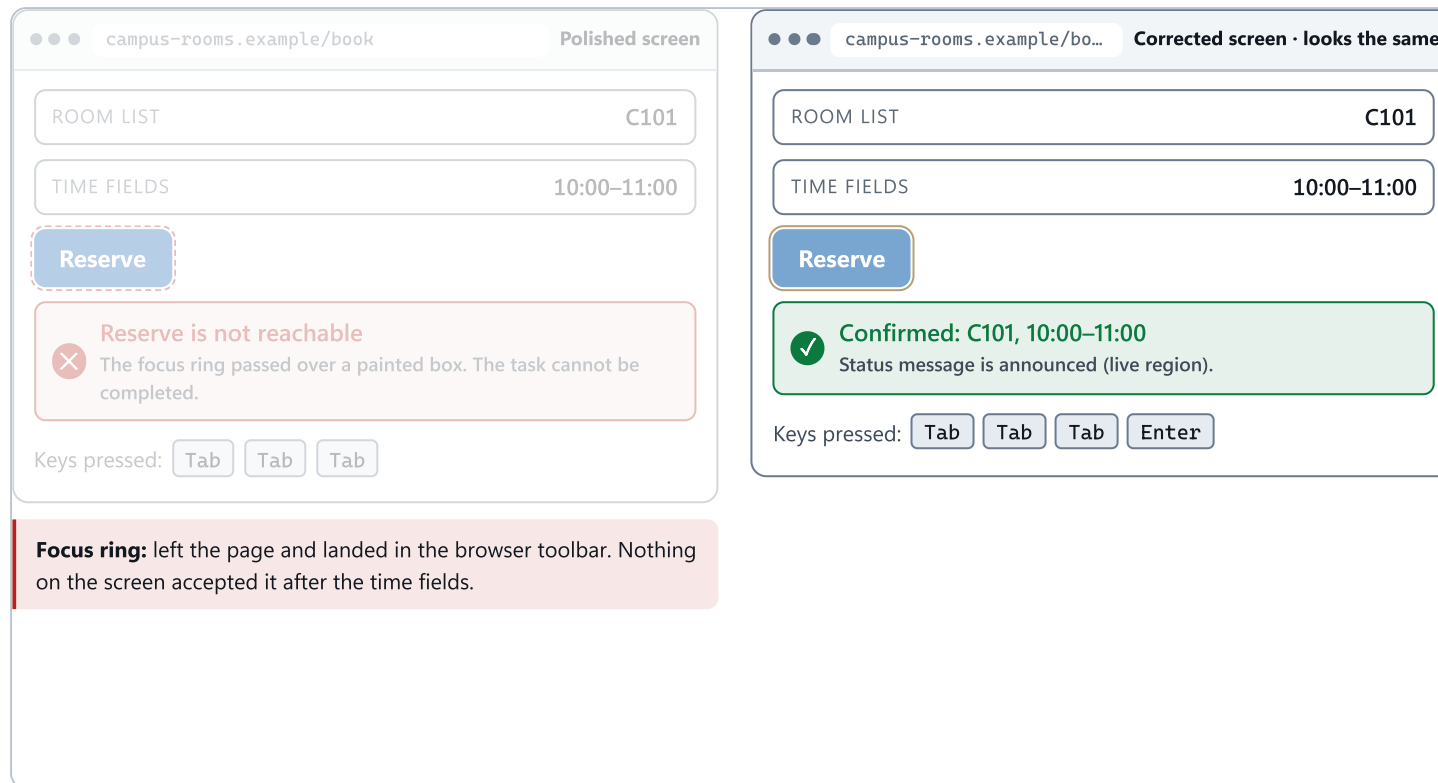
Room closures stay an open need until Week 14: who may record a closure, and what happens to affected bookings.

## The same screen can serve one person and exclude another

### Panel 2 of 2 · The keyboard route that completes the booking

Condition: **Changed condition: the same screen, used with the keyboard only**    State: 7 of 8

**Enter** → Confirmed, and the status message is announced, so a screen reader speaks it too. Rule R-05: the core task can be completed by keyboard, with status and error information you can perceive.



The failed route on the polished screen: Tab to room list; Tab to time fields; Reserve control is not reachable; task cannot be completed. The corrected route: Tab to room list; Tab to time fields; Tab to Reserve; Enter; Status message is announced. Compare on task completion, not appearance.

**Principle** · Quality depends on the people and conditions included in the definition of success.

# Faster is not a complete description of better

## Panel 1 of 2 · Two designs under the stated constraints

Condition: Baseline: response-time target: an answer within 2 s; correctness (R-01) required    State: 7 of 7

Under a 2 s target both designs are fast enough. Correctness is a condition of acceptance, so Design 1 is out before speed is compared. Faster described one attribute; better needs the stated constraints.

QUALITY ATTRIBUTES OF TWO DESIGNS (EXAMPLE ASSUMPTIONS)

| QUALITY ATTRIBUTE                                                                  | DESIGN 1 · THE SCREEN DECIDES                          | DESIGN 2 · ONE SHARED DECISION                                       |
|------------------------------------------------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------------|
| Response time until an answer appears<br><div>EXAMPLE ASSUMPTION</div>             | about 0.1 s                                            | about 0.4 s                                                          |
| Correctness when two requests overlap (R-01)<br><div>condition of acceptance</div> | ✗ two confirmations possible                           | ✓ one confirmation, one rejection                                    |
| Recoverability after a wrong confirmation                                          | ✗ someone must find and undo the double booking later  | ✓ nothing to undo                                                    |
| Privacy of other people's bookings                                                 | ✗ every screen holds a copy of the whole room calendar | ✓ the screen sends its request; only the decision reads the calendar |
| Change effort when the booking rule changes                                        | ✗ every copy of the rule in every screen changes       | ✓ one decision point changes                                         |
|                                                                                    |                                                        |                                                                      |

STATED CONSTRAINTS

Response-time target: an answer within 2 s.  
Correctness (R-01): required in every case, not traded for speed.

Example assumption: illustrative values for two fictional designs, not measurements of a real service

No winner badge: the rows are the rationale. The response-time values are example assumptions for two fictional designs.

**Principle** · Compare alternatives under stated constraints; some requirements are conditions of acceptance rather than points in a score.

# Faster is not a complete description of better

## Panel 2 of 2 · The same designs under a 0.3 s target

Condition: Changed condition: response-time target: an answer within 0.3 s; correctness (R-01) still required    State: 5 of 5

Two options keep the rule and deal with the target; the first hides a delay by giving up the rule. An optimization that preserves the rule is engineering; one that hides the delay is a return to Design 1.

### QUALITY ATTRIBUTES OF TWO DESIGNS (EXAMPLE ASSUMPTIONS)

| QUALITY ATTRIBUTE                            | DESIGN 1 · THE SCREEN DECIDES                          | DESIGN 2 · ONE SHARED DECISION                                       |
|----------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------------|
| Response time until an answer appears        | about 0.1 s                                            | about 0.4 s                                                          |
| EXAMPLE ASSUMPTION                           |                                                        |                                                                      |
| Correctness when two requests overlap (R-01) | ✗ two confirmations possible                           | ✓ one confirmation, one rejection                                    |
| condition of acceptance                      |                                                        |                                                                      |
| Recoverability after a wrong confirmation    | ✗ someone must find and undo the double booking later  | ✓ nothing to undo                                                    |
| Privacy of other people's bookings           | ✗ every screen holds a copy of the whole room calendar | ✓ the screen sends its request; only the decision reads the calendar |
| Change effort when the booking rule changes  | ✗ every copy of the rule in every screen changes       | ✓ one decision point changes                                         |

### STATED CONSTRAINTS

Response-time target: an answer within 0.3 s.  
Correctness (R-01): required in every case, not traded for speed.

### OPTIONS UNDER THE 0.3 S TARGET

| OPTION                                                                   | RULE        | EFFECT                                                         |
|--------------------------------------------------------------------------|-------------|----------------------------------------------------------------|
| Answer Confirmed before the shared decision returns                      | breaks R-01 | hides the delay and brings back two confirmations              |
| Look up the room's day faster: index the stored bookings by room and day | keeps R-01  | about 0.25 s in the example; the shared decision still decides |
| Show Checking availability, then the answer                              | keeps R-01  | the wait is visible; nobody is misled                          |

Example assumption: illustrative values for two fictional designs, not measurements of a real service

Correctness (R-01) stays required. Two options keep the rule; the first hides the delay by giving up the rule.

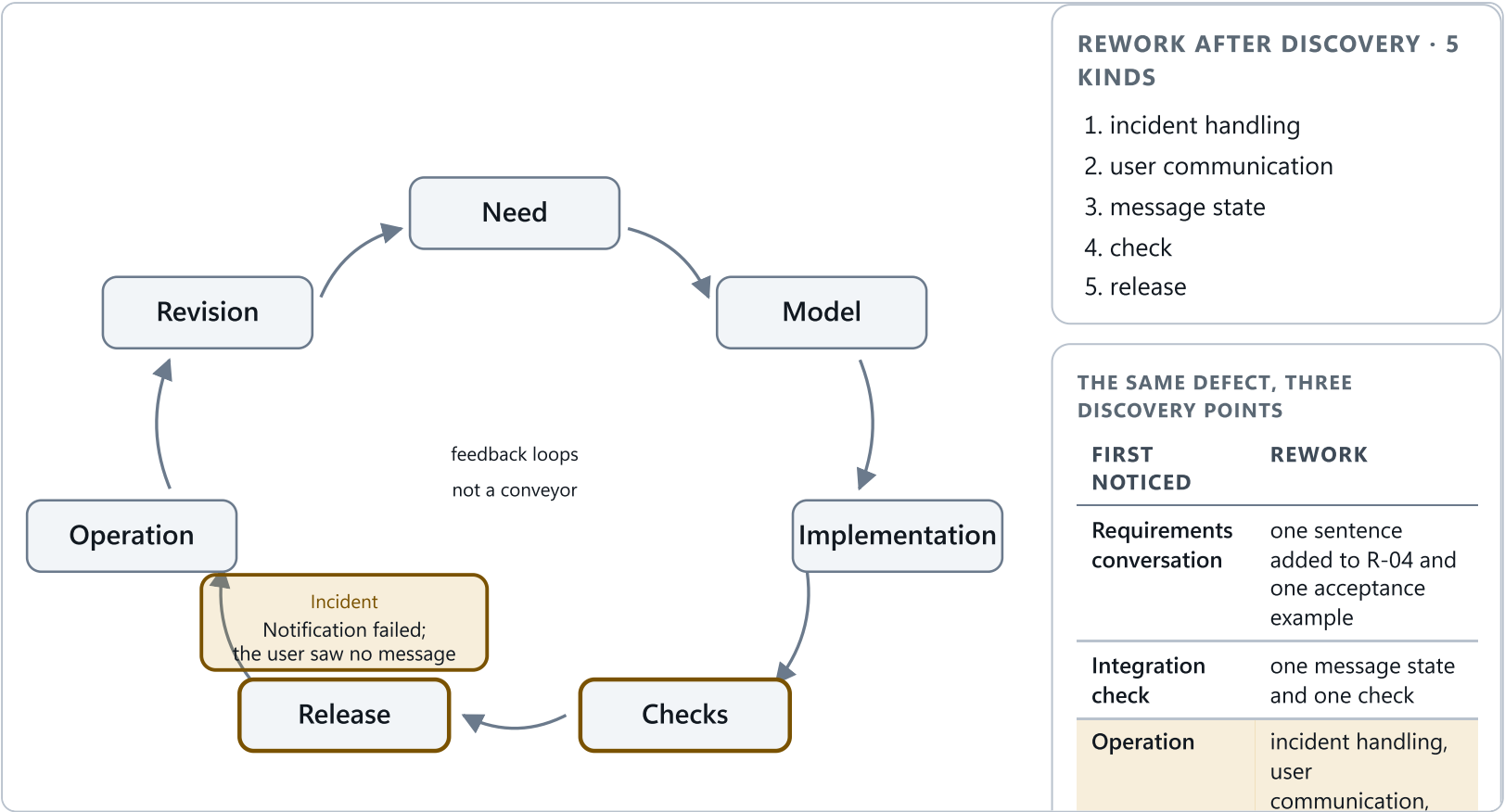
**Principle** · Compare alternatives under stated constraints; some requirements are conditions of acceptance rather than points in a score.

# A failure can teach the next requirement

## Panel 1 of 3 · The loop and the path of a defect found in operation

Condition: Baseline: the missing message is first noticed in operation, by a user    State: 6 of 7

**Release.** The change ships. Count the rework: incident handling, user communication, message state, check, release. Five kinds, and the user found the defect.



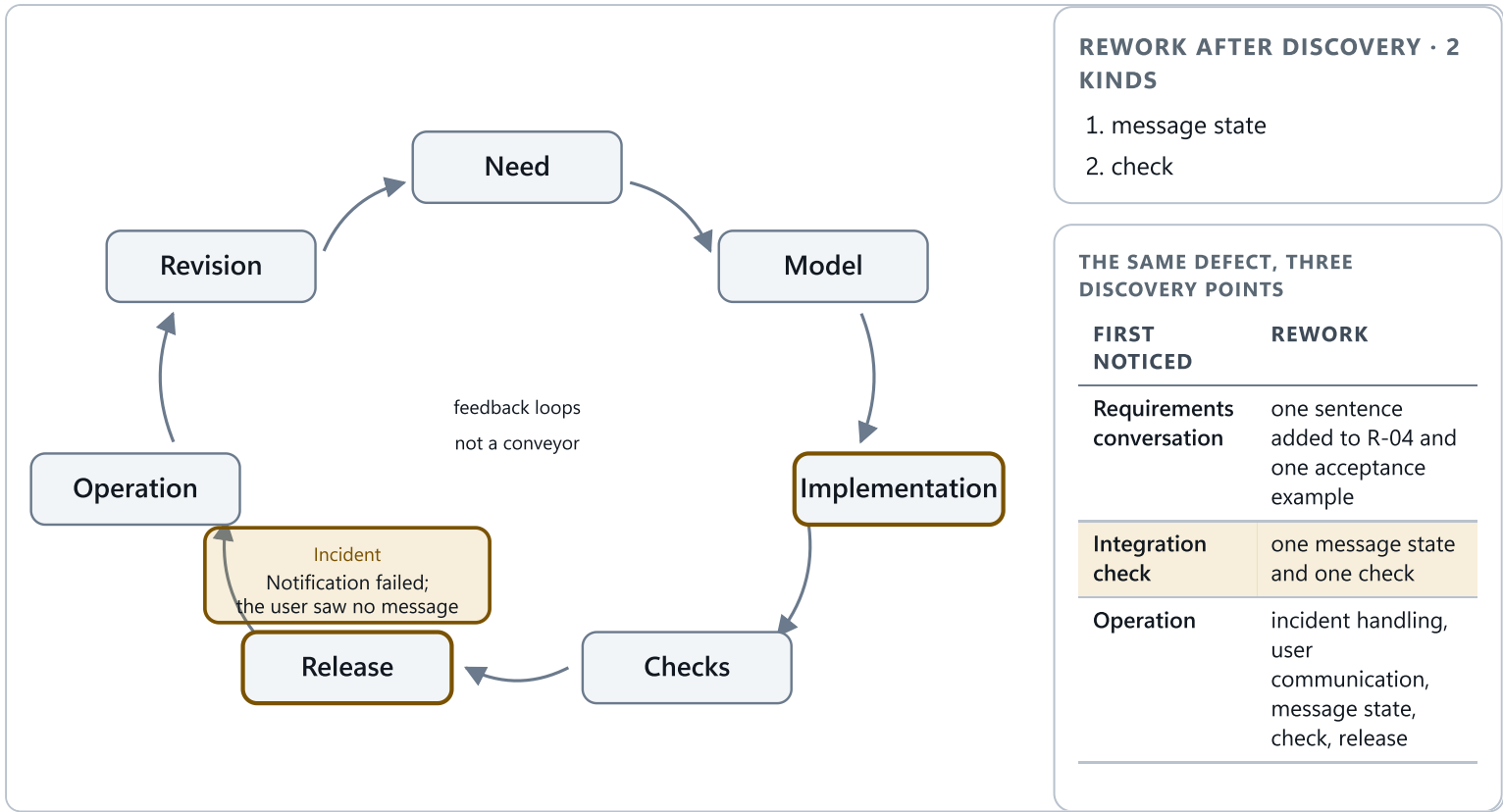
The card travelled Operation → Revision → Need and Model → Implementation and Checks → Release. Rework: incident handling, user communication, message state, check, release.

# A failure can teach the next requirement

## Panel 2 of 3 · The same defect found at an integration check

Condition: Changed condition: the same defect is first noticed at an integration check    State: 4 of 4

**Release** with the message state in place. Compare the rework list with the baseline.



Path: Checks → Model and Implementation → Release. Rework: one message state and one check. No cost multiplier is claimed; the kinds of rework are the comparison.

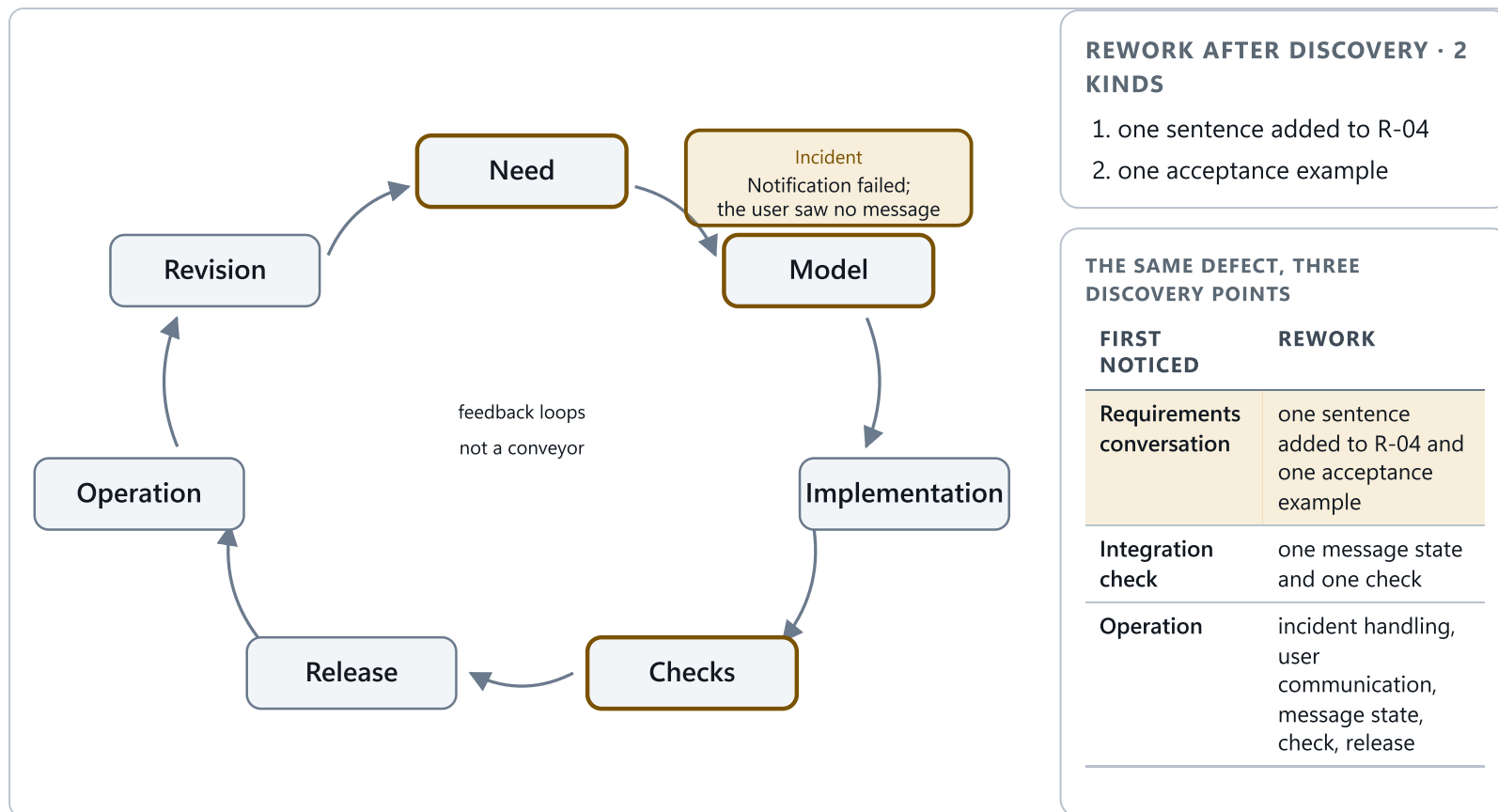
**Principle** · The lifecycle is a network of feedback loops, not a one-way conveyor ending when code runs.

## A failure can teach the next requirement

### Panel 3 of 3 · The same defect found while talking about the need

Condition: **Changed condition:** the same defect is first noticed in the requirements conversation    State: 3 of 3

The example flows into **Model** and **Checks** as they are written. Compare the three rework lists: the same defect, three different amounts of undoing.



Rework: one sentence added to R-04 and one acceptance example. Iteration revisits understanding; an increment adds usable capability.

**Principle** · The lifecycle is a network of feedback loops, not a one-way conveyor ending when code runs.

# What would make you believe this works?

## Panel 1 of 1 · Claim, check and remaining uncertainty, with the counterexample

Condition: Changed condition: workload: two people request the same room in the same moment    State: 3 of 3

The third claim named its conditions, and its checks include the two-at-once case at the storage boundary, so it covers the new workload. The habit of this course: state the evidence and its limits together.

| CLAIM                                                                                                                                     | CHECK THAT SUPPORTS IT                                                                                                    | REMAINING UNCERTAINTY                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| It looks right                                                                                                                            | ✓ Someone read the screen after one booking                                                                               | ? Nothing about the stored calendar, other people or other rooms                                                                                         |
| It passed one example                                                                                                                     | ✓ Ari books C101 10:00–11:00 on an empty calendar; the stored calendar shows one booking                                  | ? Other times, other rooms, a booking that already exists, two people at once<br>exposed assumption: the assumption that only one person books at a time |
| It met the stated checks under these conditions                                                                                           | ✓ Interval cases (overlap, adjacent, other room) and the two-at-once case at the storage boundary covers the new workload | ? Conditions outside the stated list: three people at once, a lost response, a clock that differs                                                        |
| Counterexample: Ari and Bo both request C101 10:00–11:00 in the same moment · exposes the assumption that only one person books at a time |                                                                                                                           |                                                                                                                                                          |

Counterexample: Ari and Bo both request C101 10:00–11:00 in the same moment. It exposes the assumption that only one person books at a time; the single example still holds for one person at a time.

Principle · A check supports a bounded claim; confidence without scope is difficult to evaluate.



## Who carries the cost of an engineering shortcut?

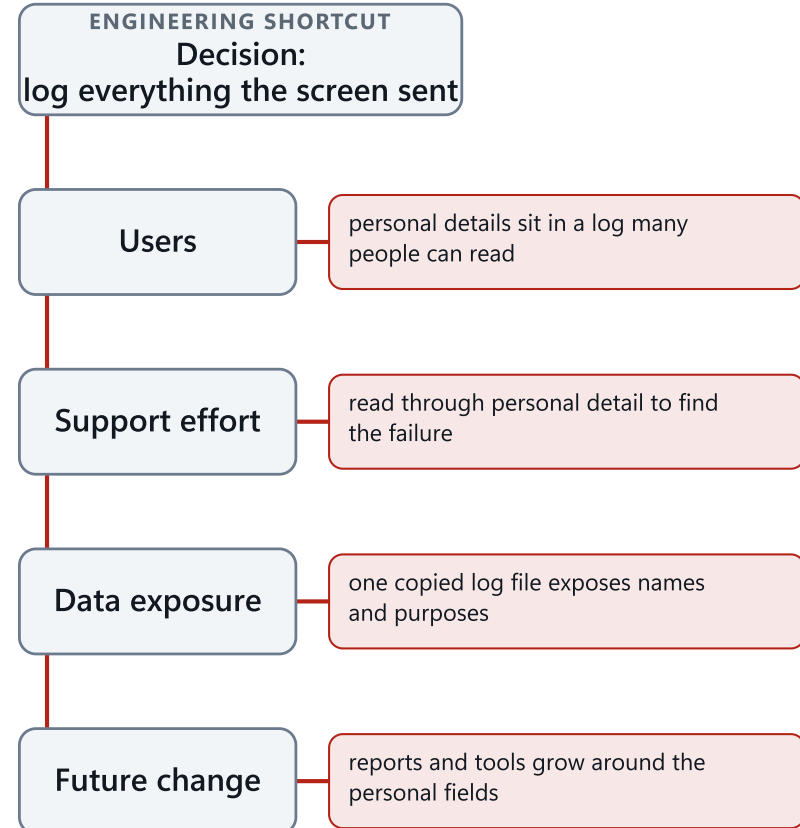
### Panel 1 of 2 · The log as first written, and the consequences it imposes

Condition: Baseline: support question: why did this booking fail? State: 5 of 7

**Future change.** Logging everything: reports and tools grow around the personal fields. Minimal record: a field is added when a question needs it.

#### SUPPORT LOG ENTRY AS FIRST WRITTEN

|               |                             |            |
|---------------|-----------------------------|------------|
| request_id    | req-4f9c                    | needed     |
| time          | 09:41:07                    | needed     |
| outcome       | failed: storage timeout     | needed     |
| user          | Ari (U-01)                  | not needed |
| room and slot | C101, 10:00-11:00           | not needed |
| purpose       | counselling session         | not needed |
| full request  | every field the screen sent | not needed |



Red: what logging everything imposes; the minimal record removes each one.

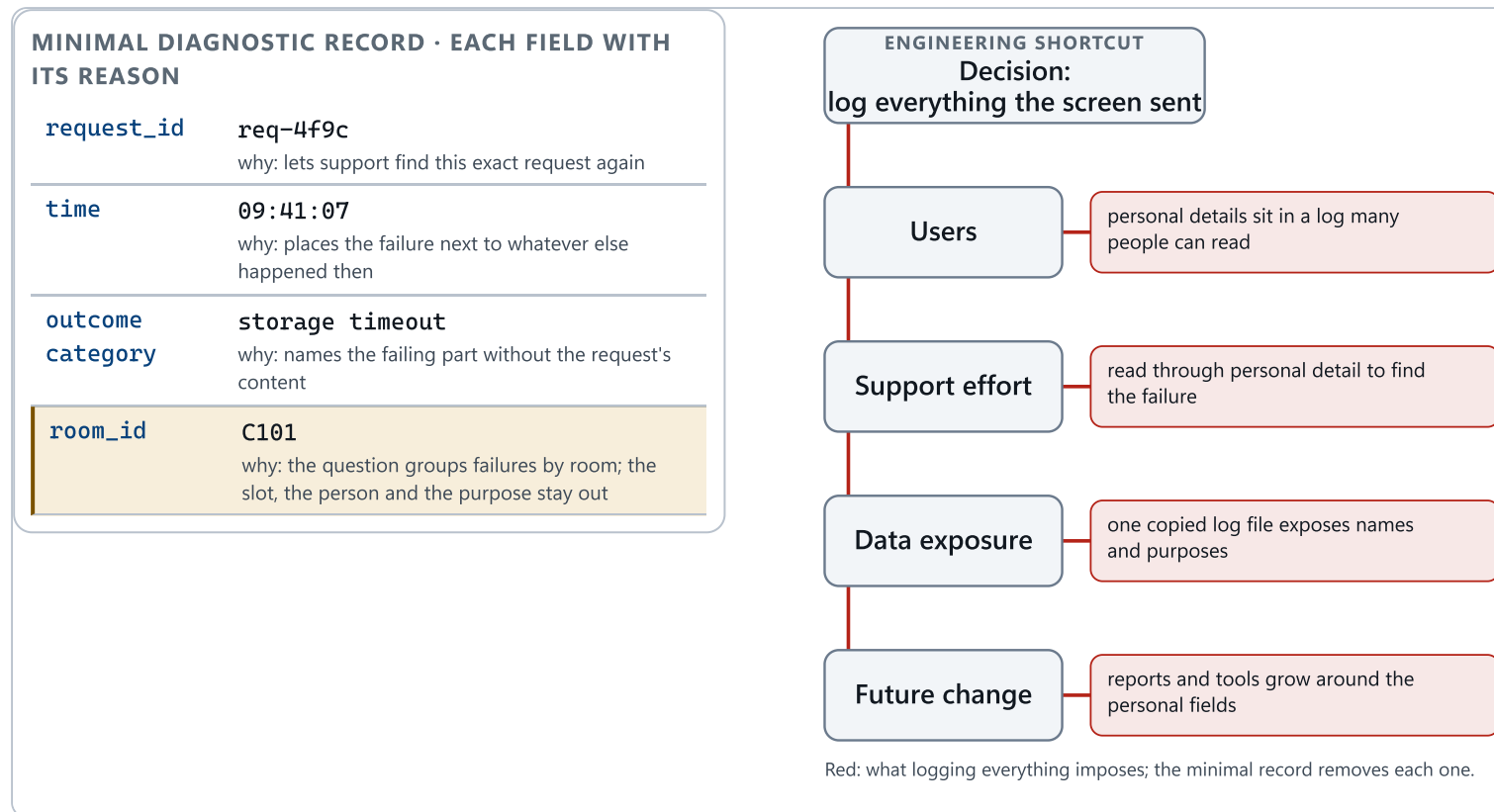
Fields marked not needed describe the person, not the failure. Each red box is a consequence for people who did not choose the logging design.

## Who carries the cost of an engineering shortcut?

### Panel 2 of 2 · The minimal diagnostic record with the rationale for each field

Condition: Changed condition: support question: is the failure specific to one room? State: 2 of 3

**room\_id = C101** is added, with its reason: the question groups failures by room; the slot, the person and the purpose stay out. Deliberate collection: one field, one written reason. Collecting everything by default needs no reason and answers no question better.



The changed support question justified exactly one more field, **room\_id**, with its reason. The slot, the person and the purpose stay out of the log.

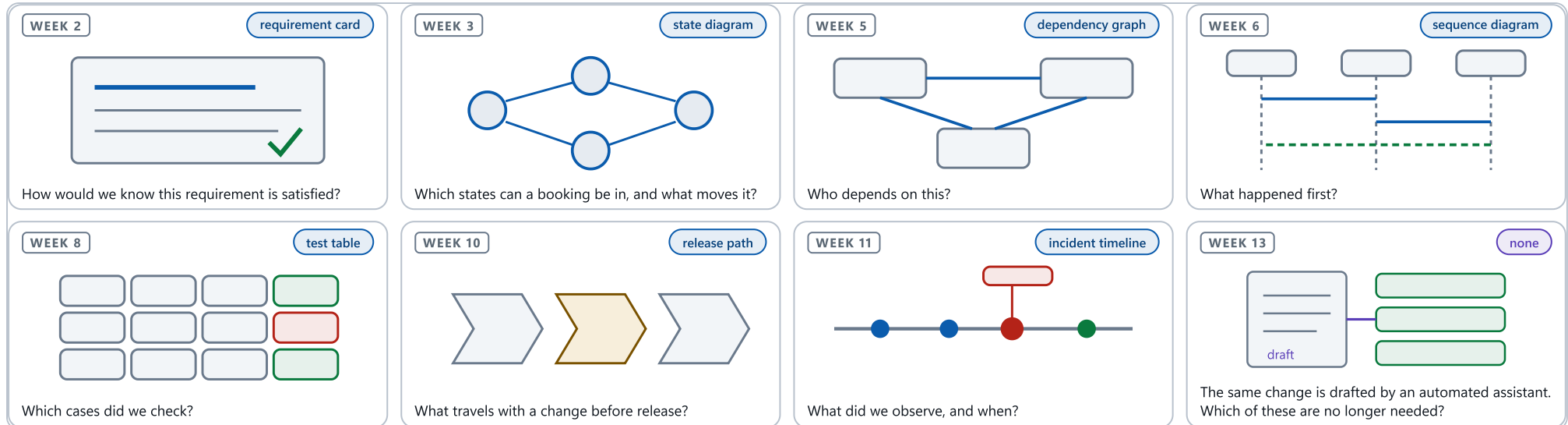
**Principle** · Professional responsibility includes the consequences imposed on people who did not make the technical choice.

# One domain, many kinds of explanation

## Panel 1 of 2 · The course map: question to representation

Condition: **Baseline:** question: who depends on the booking decision? State: 5 of 7

**Week 11:** What did we observe, and when? An incident timeline. **Week 13:** The same change is drafted by an automated assistant. Which of these are no longer needed? None: every representation and check above still applies.



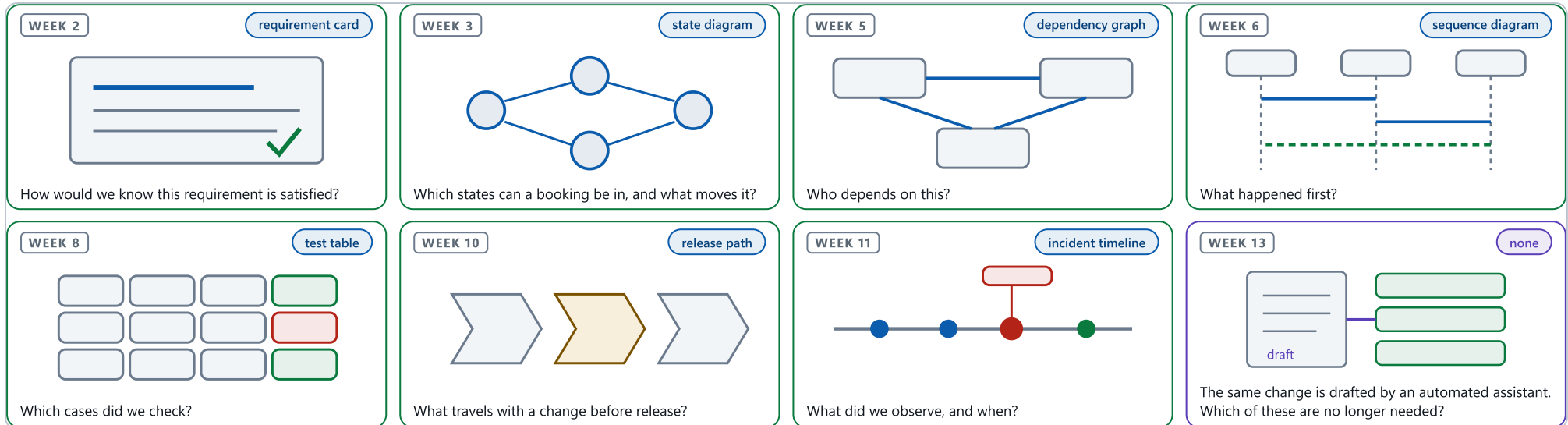
Week 2: How would we know this requirement is satisfied? → requirement card with its acceptance example. Week 3: Which states can a booking be in, and what moves it? → state diagram. Week 5: Who depends on this? → dependency graph. Week 6: What happened first? → sequence diagram. Week 8: Which cases did we check? → test table. Week 10: What travels with a change before release? → release path. Week 11: What did we observe, and when? → incident timeline. Week 13: The same change is drafted by an automated assistant. Which of these are no longer needed? → none: every representation and check above still applies.

## One domain, many kinds of explanation

### Panel 2 of 2 · The actor changes; the representations and checks stay

Condition: **Changed condition:** actor: the booking change is drafted by an automated assistant, not a colleague    State: 2 of 2

**None.** The requirement card, the state diagram, the dependency graph, the sequence, the test table, the release path and the incident timeline all still apply, because they describe the system and its behavior, not who typed the draft. Week 13 examines this in full.



A change drafted by an automated assistant still needs the requirement card, the state diagram, the dependency graph, the sequence, the test table, the release path and the incident timeline. Week 13 examines this in full.

**Principle · Choose a representation for the question, not a diagram style for the whole course.**

## Eight principles, four checks

One principle per scene. Then four questions: decide your answer before you open it.

### Principles

#### 1 · ANCHOR

Engineering judges behavior against an intended outcome and its consequences, not appearance alone.

[week-01.html#/two-confirmations](#)

#### 2 · BRIDGE

A system boundary identifies responsibility and dependencies; it does not make external behavior disappear.

[week-01.html#/system-boundary](#)

#### 3 · ANCHOR

Quality depends on the people and conditions included in the definition of success.

[week-01.html#/stakeholder-lenses](#)

#### 4 · BRIDGE

Compare alternatives under stated constraints; some requirements are conditions of acceptance rather than points in a score.

[week-01.html#/quality-tradeoffs](#)

#### 5 · ANCHOR

The lifecycle is a network of feedback loops, not a one-way conveyor ending when code runs.

[week-01.html#/feedback-lifecycle](#)

#### 6 · BRIDGE

A check supports a bounded claim; confidence without scope is difficult to evaluate.

[week-01.html#/evidence-not-confidence](#)

#### 7 · BRIDGE

Professional responsibility includes the consequences imposed on people who did not make the technical choice.

[week-01.html#/responsibility-and-impact](#)

#### 8 · BRIDGE

Choose a representation for the question, not a diagram style for the whole course.

[week-01.html#/course-evidence-map](#)

### Checks: decide your answer before reading the one below each question

#### 1. What is wrong with two successful overlapping confirmations?

They violate the intended exclusivity rule even when each screen and request appears successful.

#### 2. Why include support staff in the stakeholder map?

Their diagnostic needs influence observability, documentation, privacy, and recovery decisions.

#### 3. Does passing one booking example establish correctness for simultaneous requests?

No. The evidence does not cover that interaction condition.

#### 4. Which representation answers: what happens after a timeout?

A state model or a sequence/timeline view, depending on whether the question concerns possible states or a particular interaction.

# What carries forward, and where to read more

Next week starts from one sentence and turns it into requirements you can check.

Week 2 starts from this request:  
“Make room booking fair and easy.”

## Carried forward to Week 2

- The context map: the booking service inside its boundary, four dependencies outside.
- The no-overlap concern: one confirmed booking per room and hour (R-01), kept where bookings are stored.
- Open assumptions: who may record a room closure; what the user sees while a message is delayed; how the shared decision behaves when two requests arrive in the same instant.

## Sources for this lecture

S01

IEEE Computer Society, Guide to the Software Engineering Body of Knowledge (SWEBOK V4.0a) <https://www.computer.org/education/bodies-of-knowledge/software-engineering>

S02

ACM / IEEE-CS / AAAI, Computer Science Curricula 2023, final report <https://csed.acm.org/final-report/>

S24

Software Engineering at Google, chapter 1: What is software engineering? <https://abseil.io/resources/swe-book/html/ch01.html>

S25

W3C, Web Content Accessibility Guidelines 2.2 <https://www.w3.org/TR/WCAG22/>

## Terms used this week

Software engineering

Making and changing software that people depend on, over its whole lifetime: agreeing what it must do, building, checking, running and revising it, with the consequences for other people counted in. It is about change, coordination and quality.

Stakeholder

A person or group affected by the service: the person booking, room staff, support staff, a future maintainer.

Requirement

A statement of what the service must do or keep true, written so that a check can show whether it holds. R-01 is one.

Constraint

A condition every acceptable design must meet, such as correctness under overlapping requests.

Quality attribute

A property of how the service behaves: response time, recoverability, privacy, change effort.

Evidence

What a check showed, together with the conditions it ran under. Its scope is part of the evidence.

Lifecycle

The loop of need, model, implementation, checks, release, operation and revision that a service goes around repeatedly.

Trade-off

A choice between alternatives under stated constraints, where each option costs something the other keeps.